CHAPTER 9 REVIEW

Quiz
Language of maths 9

Puzzle
Statistics crossword

Language of maths

back-to-back	bias	categorical data	census
cluster	continuous	cumulative frequency	discrete
distribution	dot plot	histogram	mean
measure of centre	median	mode	negatively skewed
numerical data	outlier	polygon	population
positively skewed	sample	stem-and-leaf plot	symmetrical

1 What is the collective name for the mean, median and mode?

2 Is a person's arm length an example of discrete data or continuous data?

3 What is the meaning of 'skewed'?

4 Explain the difference between a *population* and a *sample*.

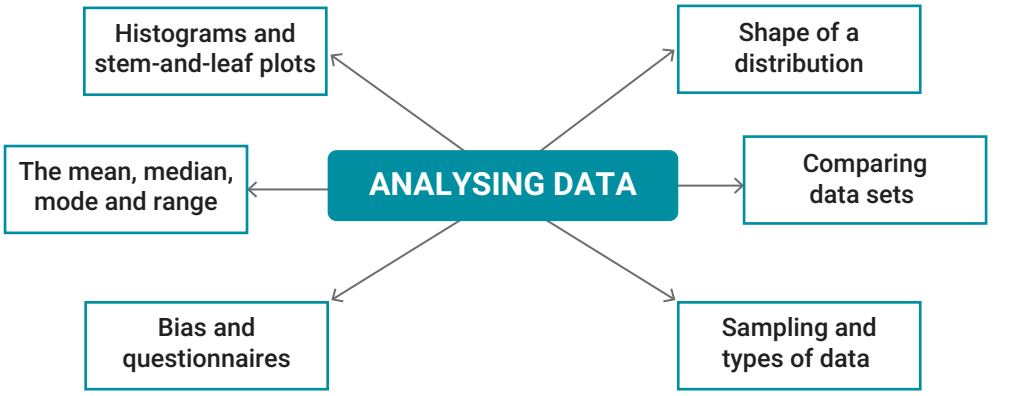
5 What name is given to an extreme value that is very different from the other values in a data set?

6 Name one way of avoiding bias when selecting a sample.

Worksheet
Mind map: Analysing data

Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 9

ANSWERS ON P. 586

Quiz
Test yourself 9

1 Find the mean, median, mode and range of this set of data. 9.01

8 6 9 4 10 8 5 5 8

2 A real estate agent sold 10 properties during 1 week. The prices of these properties are: 9.01

$420 000	$485 000	$379 000	$295 000	$1 750 000
$515 500	$315 500	$408 000	$368 000	$315 000

a Find the mean price of the properties.

b Find the median price of the properties.

c Compare the mean price with the median price. Which is the better measure of centre? Give reasons.

d When data on sales of properties are given, the median price is used, not the mean. Explain why.

3 These percentage marks were scored by 30 students in a Maths test. 9.02

45	67	78	54	22	38	58	64	73	89
84	70	78	62	56	59	42	47	86	90
64	63	72	77	43	85	75	71	70	83

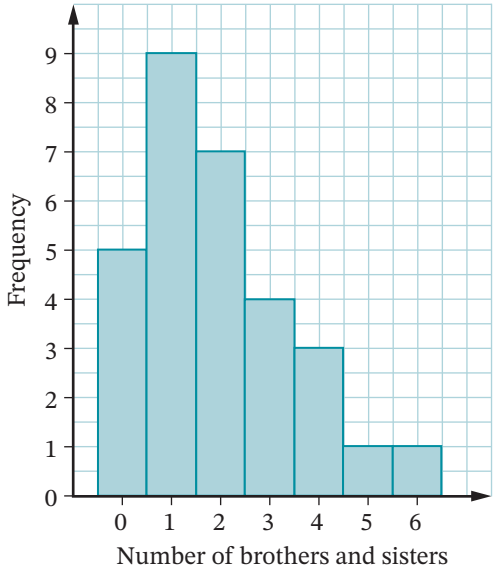
a Construct an ordered stem-and-leaf plot for the data.

b Find the mean, median, mode and range of the marks.

c Are there any outliers?

d Compare the mean and median. Which is more accurate? Give reasons.

4 A group of people were surveyed on the number of brothers and sisters they had. The results are displayed on the histogram. 9.02



a What is the range of this data?

b What is the mode?

c Find the mean correct to 2 decimal places.

d What is the median?

e Which is the best measure of centre?

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9.02

5 24 patients were chosen at random from 2 medical centres. The waiting times (in minutes) for each patient are listed below.

Northbridge:	7	16	28	27	30	30	47	9	9	15	27	11
	38	35	42	53	23	22	45	32	37	37	18	16
Westvale:	44	13	7	9	25	26	37	35	8	18	18	15
	22	24	7	5	24	24	33	37	24	15	36	43

a Display the information in a back-to-back stem-and-leaf plot.

b Find the mean, median and range of both sets of data.

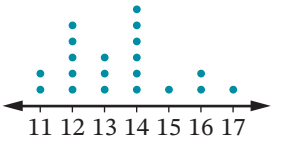
c Identify any outliers or clusters.

d Are there significant differences between the 2 sets of data? Explain your answer.

9.03

6 For each distribution, describe the shape and identify any outliers or clusters.

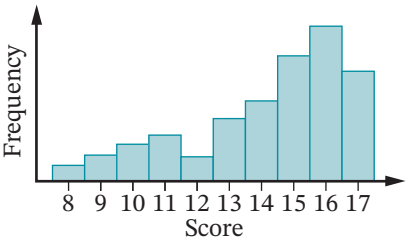
a



b

Stem	Leaf
8	7 8
9	1 3 4 4
10	0 2 7 8 8 9
11	1 2 2
12	0 7

c



9.04

7 The heights (in centimetres) of students in a Year 9 class are:

Girls:	160	153	155	156	142	163	162
	148	150	150	131	138	147	145
Boys:	150	165	164	178	170	171	164
	148	154	176	134	180	176	175

a Arrange the data in a back-to-back stem-and-leaf plot.

b Describe the shape of the display for:

i the girls **ii** the boys.

c Are there any outliers or clusters for either group?

d Find the mean (correct to 2 decimal places) and median for both groups.

e Which measure is the better indicator of the measure of centre?

f Which group is shorter?

g Which group has the greater spread of heights?

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8 Should a sample or a census be used to investigate each topic?

PATH 9.05

- **a** The most popular car colour
- **b** Whether energy-saving light globes really use less electricity
- **c** The number of retired people living on the Gold Coast, Queensland
- **d** People's views on which nation will win the soccer World Cup

9 Classify each type of data as categorical or numerical. If categorical, then classify it as nominal or ordinal. If numerical, then classify it as discrete or continuous.

9.05

- **a** The temperature in Alice Springs.
- **b** The number of views of a TikTok video.
- **c** Suburb or town of home.
- **d** The make of car passing through a motorway toll collection point.
- **e** The probability of rain today.

10 Explain why each of the following samples is biased.

PATH 9.06

- **a** A sample of online gamers were surveyed to investigate teenagers' views on road safety.
- **b** Customers in a café can give feedback about their meal by filling in a survey and posting it to the café.
- **c** At a factory, a sample of executives were surveyed about safety in the workplace.
- **d** A sample of audience members at a movie premiere were surveyed about their views on the film.
- **e** Classic Gold FM radio listeners were asked to vote for the best song of the year.

11 Explain what is wrong with each survey question and suggest an alternative.

9.06

- **a** Are you happy?
- **b** How did you enjoy your meal at our restaurant?
- **c** Australian soldiers have fought under our country's flag. Do you think that it should be changed?

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